

Markov Chains (AI HL)

ANSWER KEY



Constructing a transition matrix

- 1 A weather model has two states, Sunny (S) and Rainy (R). From Sunny: 80% stay Sunny, 20% become Rainy. From Rainy: 40% become Sunny, 60% stay Rainy. Write down the transition matrix T , using columns for the 'from' state in the order (S, R).
 T has rows (0.8, 0.4) and (0.2, 0.6) — column 1 (from Sunny) is $(0.8, 0.2)^T$ and column 2 (from Rainy) is $(0.4, 0.6)^T$, each summing to 1.

Applying the transition matrix once

- 2 Today is sunny, so $\mathbf{p}_0 = (1, 0)^T$ (order S, R). Using the matrix T from the previous question, find $\mathbf{p}_1 = T\mathbf{p}_0$, the probability distribution for tomorrow.
 $\mathbf{p}_1$ is simply column 1 of T : $\mathbf{p}_1 = (0.8, 0.2)^T$. So $P(\text{sunny tomorrow}) = 0.8$ and $P(\text{rain tomorrow}) = 0.2$.

Applying the transition matrix twice

- 3 Using $\mathbf{p}_1 = (0.8, 0.2)^T$ from the previous question, find $\mathbf{p}_2 = T\mathbf{p}_1$.
Row 1: $0.8(0.8) + 0.4(0.2) = 0.64 + 0.08 = 0.72$. Row 2: $0.2(0.8) + 0.6(0.2) = 0.16 + 0.12 = 0.28$.
So $\mathbf{p}_2 = (0.72, 0.28)^T$.

Finding the steady-state vector

- 4 Find the steady-state vector $\mathbf{s} = (s_1, s_2)^T$ for the transition matrix T above, satisfying $T\mathbf{s} = \mathbf{s}$ and $s_1 + s_2 = 1$.
 $T\mathbf{s} = \mathbf{s}$ gives $0.8s_1 + 0.4s_2 = s_1$, i.e. $0.4s_2 = 0.2s_1$. So $s_2 = 0.5s_1$. Substituting into $s_1 + s_2 = 1$ gives $s_1 + 0.5s_1 = 1$. So $s_1 = 2/3$, and $s_2 = 1/3$.

Explaining why a Markov chain is regular

- 5 Explain why the Markov chain above converges to a unique steady state regardless of the initial distribution.
Every entry of the transition matrix T is strictly positive (no transition probability is zero), so the chain is regular. A regular chain's distribution $\mathbf{p}_n$ converges to the same steady-state vector $\mathbf{s}$ as $n \rightarrow \infty$, whatever the starting distribution $\mathbf{p}_0$ is.

Computing and interpreting T^2

- 6 Using the transition matrix T with rows $(0.8, 0.4)$ and $(0.2, 0.6)$, find T^2 and interpret its $(1,1)$ entry. T^2 has entries: $(1, 1) = 0.8(0.8) + 0.4(0.2) = 0.72$; $(1, 2) = 0.8(0.4) + 0.4(0.6) = 0.56$; $(2, 1) = 0.2(0.8) + 0.6(0.2) = 0.28$; $(2, 2) = 0.2(0.4) + 0.6(0.6) = 0.44$. The $(1, 1)$ entry, 0.72, is the probability of being Sunny two days from now given it is Sunny today — matching $\mathbf{p}_2$'s first entry found earlier.
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Constructing a 3-state transition matrix

- 7 A game has three states: Win (W), Draw (D), Loss (L). From W: 50% stay W, 30% go to D, 20% go to L. From D: 40% go to W, 20% stay D, 40% go to L. From L: 10% go to W, 30% go to D, 60% stay L. Write down the transition matrix T , using columns for the 'from' state in the order (W, D, L). T has rows (in order W, D, L): row W = $(0.5, 0.4, 0.1)$; row D = $(0.3, 0.2, 0.3)$; row L = $(0.2, 0.4, 0.6)$. Each column sums to 1, e.g. column W: $0.5 + 0.3 + 0.2 = 1$.
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Applying the transition matrix to a 3-state chain

- 8 Today's state is Draw, so $\mathbf{p}_0 = (0, 1, 0)^T$ (order W, D, L). Using the matrix T above, find $\mathbf{p}_1 = T\mathbf{p}_0$. $\mathbf{p}_1$ is column D of T : $\mathbf{p}_1 = (0.4, 0.2, 0.4)^T$. So $P(W) = 0.4$, $P(D) = 0.2$, $P(L) = 0.4$.
- 9 Using $\mathbf{p}_1 = (0.4, 0.2, 0.4)^T$ from the previous question, find $\mathbf{p}_2 = T\mathbf{p}_1$.
Row W: $0.5(0.4) + 0.4(0.2) + 0.1(0.4) = 0.2 + 0.08 + 0.04 = 0.32$. Row D:
 $0.3(0.4) + 0.2(0.2) + 0.3(0.4) = 0.12 + 0.04 + 0.12 = 0.28$. Row L:
 $0.2(0.4) + 0.4(0.2) + 0.6(0.4) = 0.08 + 0.08 + 0.24 = 0.40$. So $\mathbf{p}_2 = (0.32, 0.28, 0.40)^T$ (entries sum to 1.00, as expected).
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Defining the steady-state equation

- 10 State the equation that defines a steady-state vector $\mathbf{s}$ of a transition matrix T , and explain what it means for $\mathbf{s}$ to be an eigenvector of T with eigenvalue 1.
A steady-state vector satisfies $T\mathbf{s} = \mathbf{s}$, with the entries of $\mathbf{s}$ summing to 1. Saying $\mathbf{s}$ is an eigenvector of T with eigenvalue 1 means exactly this: applying the transition matrix to $\mathbf{s}$ leaves it completely unchanged, so once the distribution reaches $\mathbf{s}$ it stays there forever.
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Verifying a steady-state vector directly

- 11 Verify that $\mathbf{s} = (2/3, 1/3)^T$ is the steady-state vector for the weather transition matrix T above, by checking $T\mathbf{s} = \mathbf{s}$ directly.
- Row 1: $0.8(2/3) + 0.4(1/3) = 1.6/3 + 0.4/3 = 2/3$, matching s_1 . Row 2:
 $0.2(2/3) + 0.6(1/3) = 0.4/3 + 0.6/3 = 1/3$, matching s_2 . Since $T\mathbf{s} = \mathbf{s}$ and $2/3 + 1/3 = 1$, $\mathbf{s}$ is confirmed as the steady-state vector.
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The memoryless (Markov) property

- 12 Explain what is meant by saying a Markov chain has the 'memoryless' property.
- The probability of moving to any state at the next step depends only on the current state, not on the sequence of states that preceded it — the history of how the system arrived at its current state has no further effect on predicting its future.